

Experiment 'p_13_9_recursive_sum_seq' Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_13_9_recursive_sum_seq

Natural language statement: Write a recursive method that takes an array and returns the sum of the values.

Method signature: p_13_9_recursive_sum_seq(inputs: seq<int>) returns (sum: int)

Ensures

- `sum == sum_array(inputs)`

Functional Code Given

```
function sum_array(inputs: seq<int>): int
  ensures sum_array(inputs) == (if |inputs| == 0 then 0 else inputs[0] + sum_array(inputs
    ↪ [1..]))
{
  if |inputs| == 0 then 0
  else inputs[0] + sum_array(inputs[1..])
}
```

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a recursive method that takes an array and returns the sum of the values.

The signature should be:

`p_13_9_recursive_sum_seq(inputs: seq<int>) returns (sum: int)`

The method should respect the following contract:

`ensures sum == sum_array(inputs)`

The contract uses the following dafny code:

```
function sum_array(inputs: seq<int>): int
  ensures sum_array(inputs) == (if |inputs| == 0 then 0 else inputs[0] + sum_array(inputs
    ↪ [1..]))
{
  if |inputs| == 0 then 0
  else inputs[0] + sum_array(inputs[1..])
}
```

These function(s) must not be used in your implementation of the method.

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnAqKOv5n4h4704dYPEXhpmU2Loto

Model: gpt-5.5-2026-04-23

Created at: 1780612124

Finish reason: stop

Usage: 448 tokens in, and 474 tokens out

```
{
  if |inputs| == 0 {
    sum := 0;
  } else {
    var tailSum := p_13_9_recursive_sum_seq(inputs[1..]);
    sum := inputs[0] + tailSum;
  }
}
```

Final Program

```
function sum_array(inputs: seq<int>): int
  ensures sum_array(inputs) == (if |inputs| == 0 then 0 else inputs[0] + sum_array(inputs
    ↪ [1..]))
{
  if |inputs| == 0 then 0
  else inputs[0] + sum_array(inputs[1..])
}

method p_13_9_recursive_sum_seq(inputs: seq<int>) returns (sum: int)
  ensures sum == sum_array(inputs)
{
  if |inputs| == 0 {
    sum := 0;
  } else {
    var tailSum := p_13_9_recursive_sum_seq(inputs[1..]);
    sum := inputs[0] + tailSum;
  }
}
```

Total Token Usage

Input tokens: 448

Output tokens: 474

Reasoning tokens: 397

Sum of 'total tokens': 922

Experiment Timings

Overall Experiment started at 1780612123395, ended at 1780612147669, lasting 24274ms (24.27 seconds)

Iteration #1 started at 1780612123395, ended at 1780612147668, lasting 24273ms (24.27 seconds)